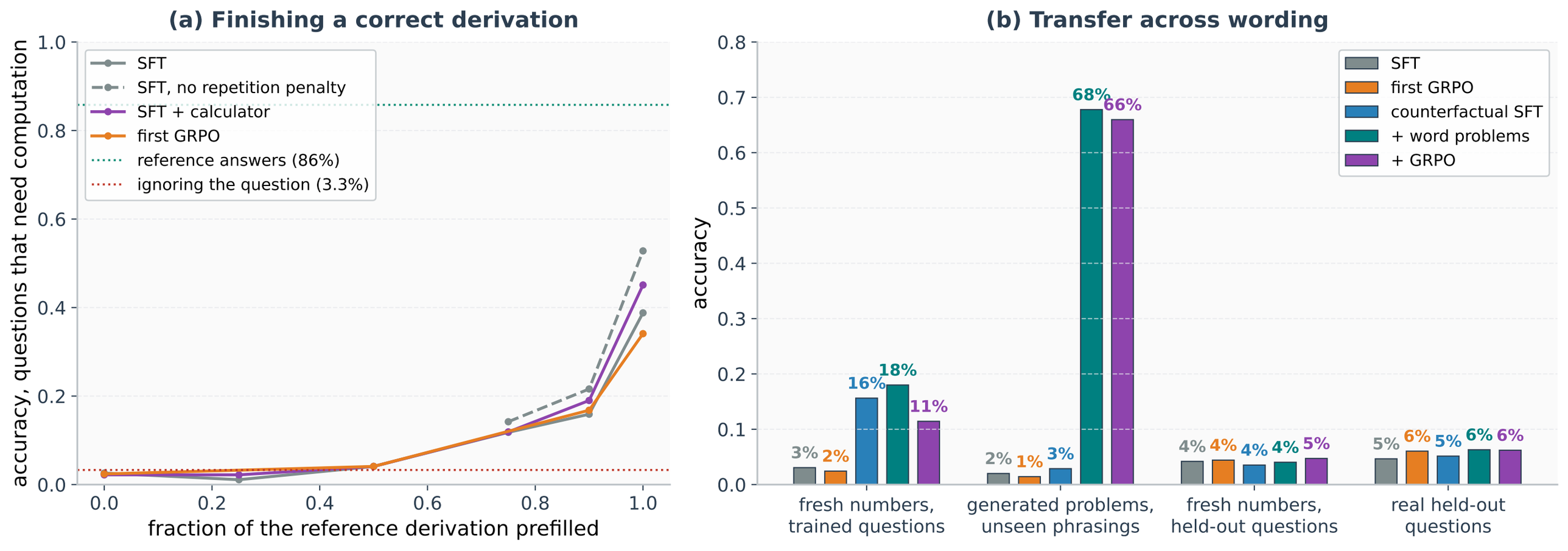
(a) Finishing a correct derivation
accuracy, questions that need computation
fraction of the reference derivation prefilled
SFT
SFT, no repetition penalty
SFT + calculator
first GRPO
reference answers (86%)
ignoring the question (3.3%)
0.0
0.2
0.4
0.6
0.8
1.0
(b) Transfer across wording
accuracy
SFT
first GRPO
counterfactual SFT
+ word problems
+ GRPO
3%
2%
16%
18%
11%
2%
1%
3%
68%
66%
4%
4%
4%
4%
5%
5%
6%
5%
6%
6%
fresh numbers, trained questions
generated problems, unseen phrasings
fresh numbers, held-out questions
real held-out questions